

AIMO Proof Pilot — Final Grade

Problem (#_ID): 3_INDEX

Submission: model_5

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

A complete, correct solution following the official approach. It introduces the encoding polynomial, gives the cyclotomic factorisations and establishes their coprimality (distinct cyclotomic factors), rules out index ≥ 6 , and exhibits a multiset of index 5 with minimum element 16, yielding $N = 5$ and $M = 16$. The only candidate deduction is that the solution does not explicitly state why the coefficients a, b, c in the final factorisation are integers — integrality being what would sharpen a bound of the form $a > 0$ to $a \geq 1$ (the point **NA** raised as **ME2**). However, the solution proves the *stronger* relations $b = 2a$ and $c = a$; at that point a is manifestly an integer, since the quotient is then an integer multiple of a monic polynomial and its leading coefficient forces $a \in \mathbb{Z}$. The integrality is not invoked before this stronger fact is available, so the omission does not reach the threshold for **ME2** and is not penalised. Scores **7**.